ASL Aspire Redesign

Enhancing Accessibility, Personalization, and Motivation

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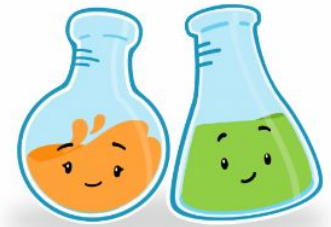
Context

- ASL Aspire is a STEM learning platform for **deaf and hard-of-hearing students**.
- Uses **ASL videos**, **visuals**, and **interactive activities** to make subjects **more accessible**.
  - Goals:
    - support **DHH learners**
    - promote conceptual understanding through **visual and signed representations**
    - help teachers guide **student progress**.

ASL ASPIRE

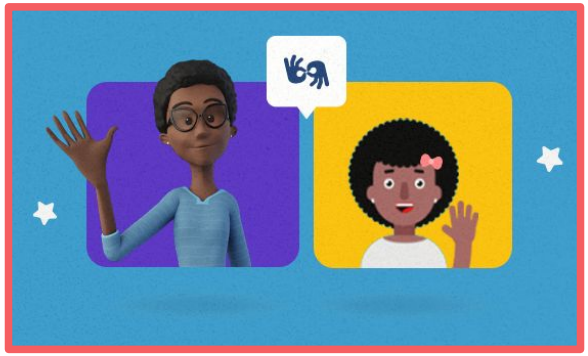
Identified Challenges

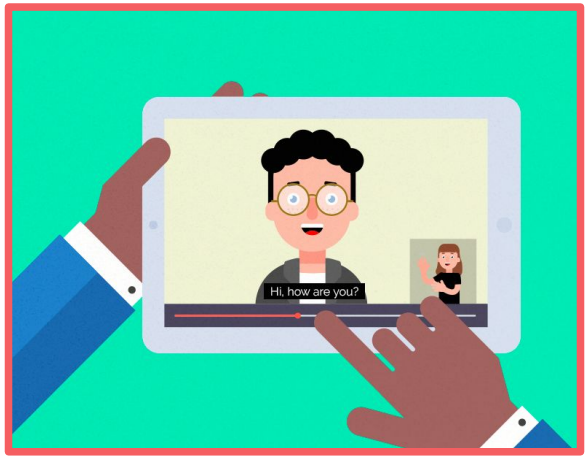
- **Only ASL + English supported**
  - (multilingual learners lack access)
- **Limited pacing/scaffolding controls** for teachers and students
- **Low motivation** and **unclear progress**
  - Students struggle to see their progress and stay motivated
- **No personalized pathways**



Research

- Academic work on multilingual DHH learners highlights the need for **multiple language pathways** and **flexible representations**.
- Research on adaptive learning shows that personalized pacing and scaffolded supports **improve engagement and comprehension**.
- Learning sciences like UDL emphasize offering multiple means of **representation, engagement,** and **expression**.





Design Focus

- **Multilingual accessibility**
  - multiple sign and written languages
- **Adaptive pacing & scaffolding**
  - step-by-step directions, hints, difficulty controls
- **Motivation: mastery + dashboard**
  - mastery levels, badges, and a redesigned dashboard
- **Personalized lessons**
  - recommendations based on mastery and performance

Interview

Interview Insights

We interviewed **Mona**, the CEO of ASL Aspire, who has led the platform's development for 5 years

Example questions:

1. What challenges do multilingual deaf learners currently face when using ASL Aspire, and what makes multilingual support difficult to implement?

2. Where do students typically get stuck during lessons or games, and what kinds of scaffolding or pacing controls would best support them?

3. Which of our proposed features—such as a languages tab, adaptive pacing, or a redesigned dashboard—best align with ASL Aspire's long-term vision?

Interview - Key Takeaways

- **Multilingual support** is a **goal**, but they want to fully flesh out a year's worth of content first.
- They want to add **leaderboards** to **boost motivation**.
- They want a **mastery level system** that makes progress clearer.
- They want **personalized learning pathways** for each student.

More Specific Interview Feedback

1. Multilingual Support: Vision but Limited by Resources
2. Student Feedback → Major Product Shifts
3. Motivation is Key: Badges & Competition
4. Current Priority: Content, Especially Math
5. Desired Future Features
6. UDL & Accessibility Considerations

Design Plan Refinements

Based on feedback:

- We added a **leaderboard** feature
- We added a **tiered badge mastery system**
- We **improved the dashboard** to **emphasize progress and goals**
- We built a **personalized lessons panel** tied to mastery

Prototype Demo

Prototype Highlights

Our prototype illustrates a rough representation of how we intend a finished redesign to look like:

- Visual **dashboard**
- Languages screen with separate **sign + written options**
- Personalized **lesson recommendations**
- Mastery **levels and badges**
- Leaderboard
- Adaptive pacing

Accessibility

Our design aligns with:

- **UDL principles**
  - Multiple means of representation (sign languages, text complexity controls)
  - Multiple means of engagement (badges, leaderboard, adaptive pacing)
- **Multimodal learning**
  - sign videos + visuals + text
- **Accessibility features**
  - high contrast mode, reduced animations, adjustable model size

Conclusion

We believe that our redesign improves:

- **Access** for **multilingual DHH learners**
- **Personalization** for **diverse learning needs**
- **Motivation** through **mastery and competition**
- **Clarity** through **better progress tracking**

Next steps…

- Refine prototype
- Run our final prototype through Mona